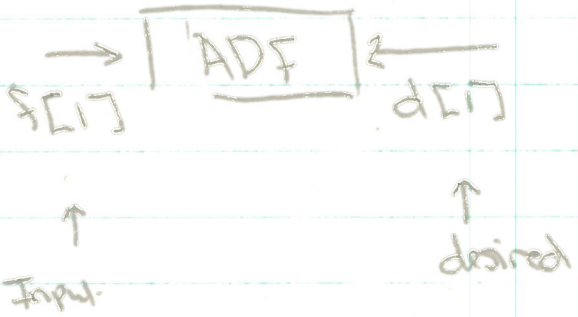
03/01/23 Signals

I

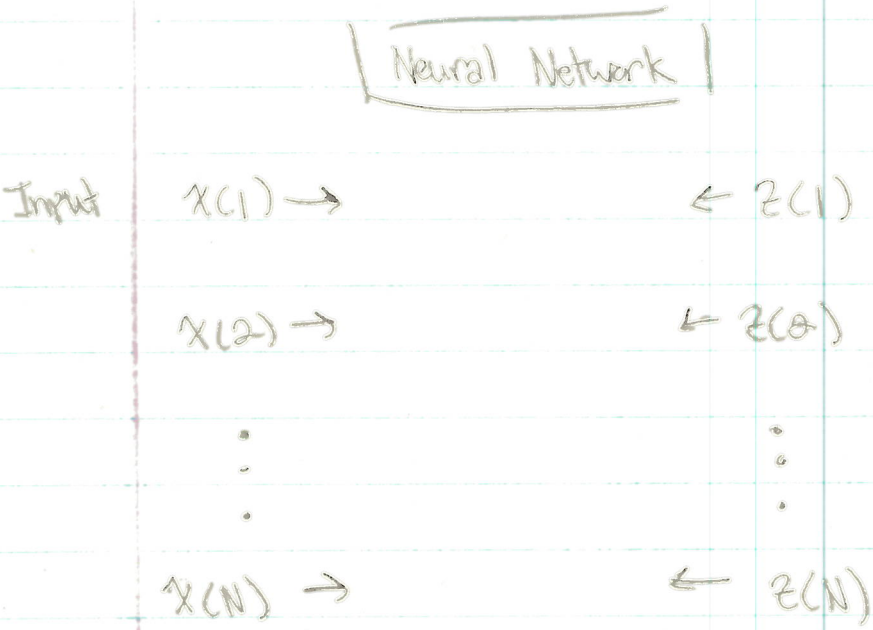
Don't worry about too much

Neural Networks

ADF: Adaptive Filter

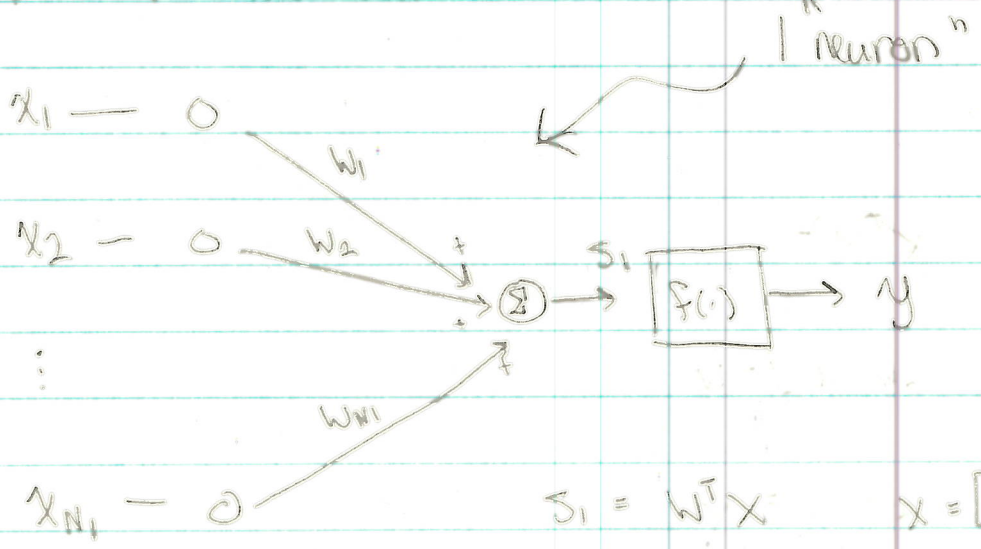


A)



It is just like an adaptive filter, but you have a fixed set of training data, the inputs and outputs can be vectors, and "Neural Network" box is nonlinear.

B) "Open the Box"

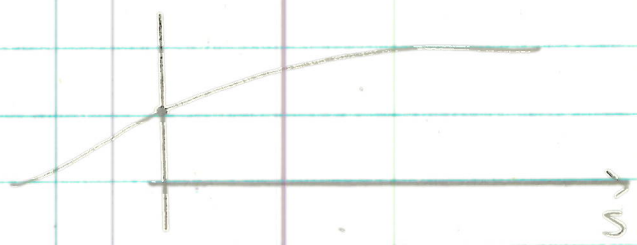


$s_1 = w^T x$ $\quad x = [x_1 \; x_2 \cdots x_{N_1}]^T$

$w = [w_1 \; w_2 \cdots w_{N_1}]^T$

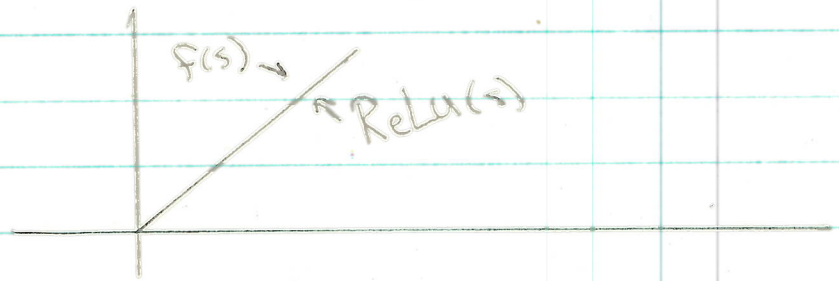
The function $f(\cdot)$ may vary, but a popular one was

$$f(s) = \frac{1}{1+e^{-s}}$$

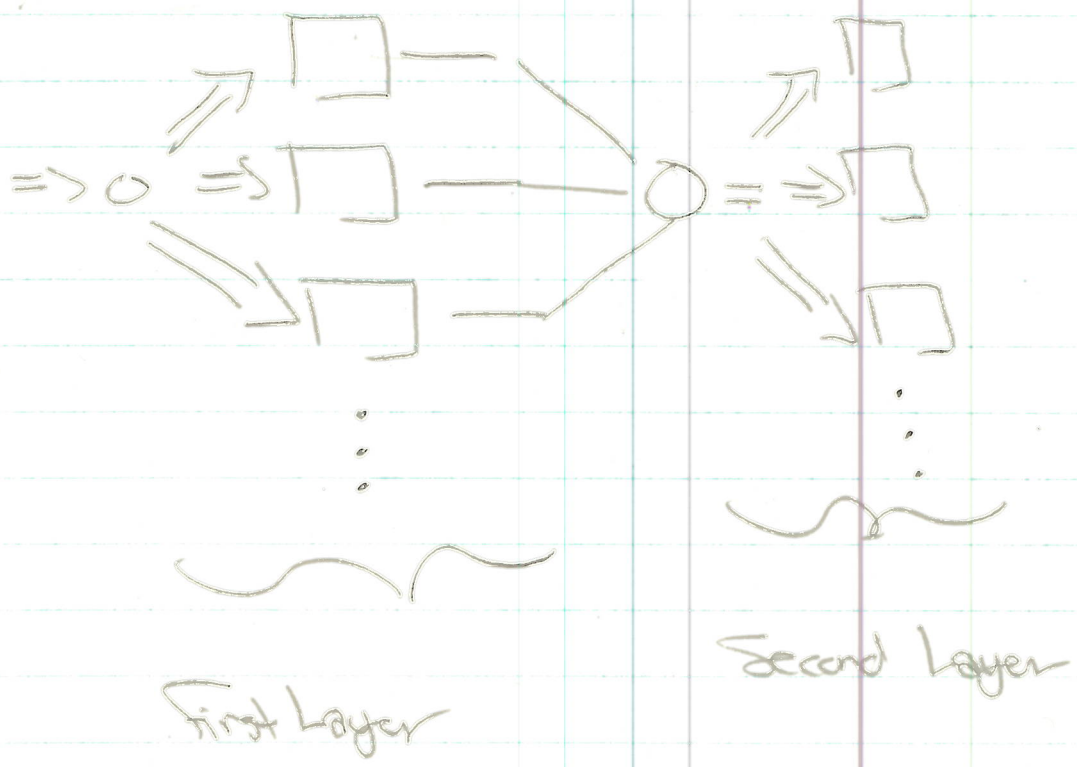


More recently what has been used is

$$\mathrm{ReLu}(s) = \begin{cases} 0 & \text{if } x < 0 \\ x & \text{if } x \geq 0 \end{cases}$$



c) Connect the neurons

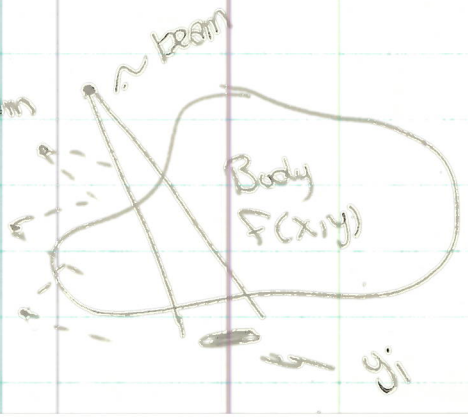


So this is denoted as a two layer neural network. This assembly is what goes in the box.

II Natural Pixel Tomography

Pump energy into a body, collect on the data, stack it, and create an image.

$$\phi(x,y) = \begin{cases} 1 & \text{if } (x,y) \text{ is in the beam} \\ 0 & \text{otherwise} \end{cases}$$



This is like an inner product

$$y_i = \iint_{\Omega} f(x,y)\,\phi_i(x,y)\,dx\,dy + \underbrace{V_i}_{\text{noise}}$$

So this integrates the function $f(x,y)$ over the beam. We then sweep the beams about the body

$$\underline{y} = [y_1 \; y_2 \; \cdots \; y_n]$$

But how to we estimate $f(x,y)$ to recreate an image out of $\underline{y}$?

A) How to "solve" this function

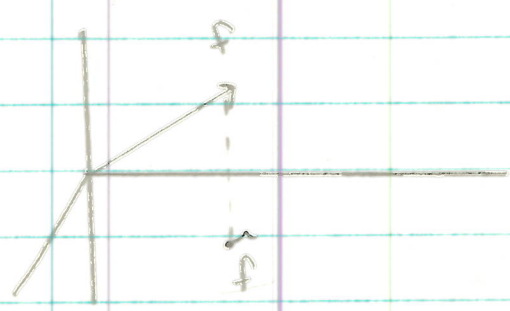
Let

$$\underbrace{\hat{f}(x,y)}_{\text{our estimat of the } f(x,y)} = \sum_{i=1}^{N} \hat{x}_i \underbrace{\phi_i(x,y)}_{\text{Basis function}}$$

↳ The beam functions

1) How to get $\hat{x}_i$?

$$\min \|f - \hat{f}\|_2^2$$



Then pick the norm

$$\iint (f(x,y) - \hat{f}(x,y))^2 \, dx\,dy$$

- The define the inner product
- $\langle a(x,y), b(x,y) \rangle = \iint a(x,y)\, b(x,y)\, dx\,dy$
- error $\perp$ data

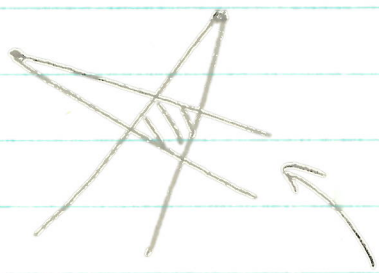
Which then produces the Gram Matrix

$$\begin{bmatrix} \langle \phi_1, \phi_2 \rangle & \langle \phi_1, \phi_2 \rangle & \cdots & \langle \phi_1, \phi_N \rangle \\ \langle \phi_2, \phi_1 \rangle & \langle \phi_2, \phi_2 \rangle & \cdots & \langle \phi_2, \phi_N \rangle \\ & \vdots & & \\ & & & \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_N \end{bmatrix} = \begin{bmatrix} \langle f, \phi_1 \rangle \\ \langle f, \phi_2 \rangle \\ \vdots \\ \langle f, \phi_N \rangle \end{bmatrix}$$

$$\langle \phi_i, \phi_j \rangle = \iint \phi_i(x,y)\, \phi_j(x,y)\, dx\,dy$$

What does this mean?

Lets draw a picture!



The cross product will produce the area of the intersection!

What do we know about $\langle f, \phi_i \rangle$? This our measurement of the density of the beam! In other words, its our measurement. Now we can solve for the coefficients x_i. Wow :o!